

AWS QUICK-LOAN PROJECT

Aim- AWS Quick-Loan Project is a user-friendly web-based loan application system that allows customers to explore loan products and submit loan applications online, with applicant details stored in a database.

Summary: Quick Loan is a web-based loan application system that provides information about Home, Gold, Vehicle, and Personal Loans. Users can submit an online application by entering their personal details and selecting a loan type. The application is processed using PHP and the submitted data is stored in a MySQL database. The system is deployed using an Nginx web server.

Work Flow:

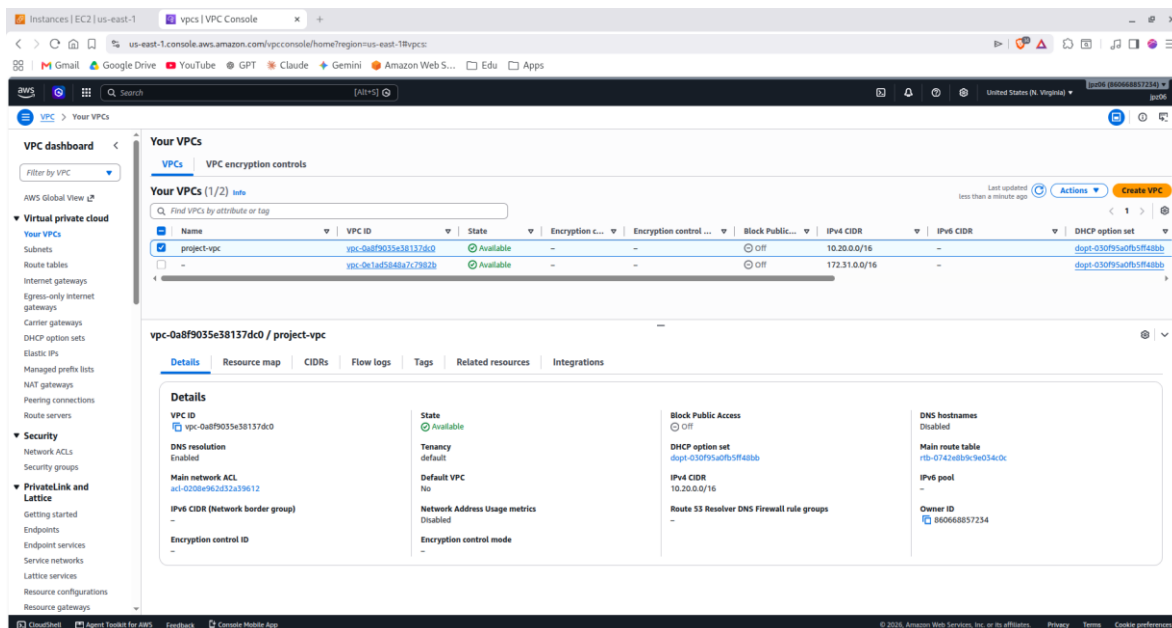
=> User → Quick Loan Website → Loan Application Form → PHP Backend → MySQL Database → Application Stored → Confirmation/Redirect

=====

Steps:

Virtual Private Cloud (VPC)

=> Create a VPC, named it as 'project-vpc'



Subnets:

Create 4 Subnets (3 public) and 1 private

Subnets (4/15) Info							
Find subnets by attribute or tag							
☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	
☐	-	subnet-0ec9a2d3b5771aeea	Available	vpc-0e1ad5848a7c7982b	Off	172.31.64.0/20	
☐	-	subnet-097129eb24fc2ffc7	Available	vpc-0e1ad5848a7c7982b	Off	172.31.0.0/20	
☐	-	subnet-084c2e95b32a7ebc4	Available	vpc-0e1ad5848a7c7982b	Off	172.31.80.0/20	
☐	-	subnet-0a83d5741923dfe25	Available	vpc-0e1ad5848a7c7982b	Off	172.31.32.0/20	
☐	-	subnet-03117be5deaf15b6e	Available	vpc-0e1ad5848a7c7982b	Off	172.31.16.0/20	
☐	-	subnet-058242e4ce721e8e8	Available	vpc-0e1ad5848a7c7982b	Off	172.31.48.0/20	
☒	private-subnet	subnet-054397ed3c1e048ad	Available	vpc-0a8f9035e38137dc0 proj...	Off	10.20.0.0/24	
☒	public-jump-subnet	subnet-0e8294c077be3d0d2	Available	vpc-0a8f9035e38137dc0 proj...	Off	10.20.1.0/24	
☒	public-web1-subnet	subnet-02d3639adc4f668d3	Available	vpc-0a8f9035e38137dc0 proj...	Off	10.20.3.0/24	
☒	public-web2-subnet	subnet-03cbe4f9256aedfdb	Available	vpc-0a8f9035e38137dc0 proj...	Off	10.20.2.0/24	

Route Tables.

1. Public Route Table

rtb-09f0556615ef05437 / public-rt [Actions](#)

Details [Info](#)

Route table ID
[rtb-09f0556615ef05437](#)

VPC
[vpc-0a8f9035e38137dc0](#) | [project-vpc](#)

Main
☒ No

Owner ID
[860668857234](#)

Explicit subnet associations
3 subnets

[subnet-02d3639adc4f668d3](#) / [public-web1-subnet](#)
[subnet-03cbe4f9256aedfdb](#) / [public-web2-subnet](#)
[subnet-0e8294c077be3d0d2](#) / [public-jump-subnet](#)

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0658f6bcb7b06589	Active	No	Create Route
10.20.0.0/16	local	Active	No	Create Route Table

Both [Edit routes](#)

=> Connect Public Route Tables with Public Subnets.

2. Private Route Table

rtb-0fb557c64a02a92a2 / private-rt [Actions](#)

Details [Info](#)

Route table ID
[rtb-0fb557c64a02a92a2](#)

VPC
[vpc-0a8f9035e38137dc0](#) | [project-vpc](#)

Main
☒ No

Owner ID
[860668857234](#)

Explicit subnet associations
[subnet-054397ed3c1e048ad](#) / [private-subnet](#)

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0f8f154c46bfa6bc1	Active	No	Create Route
10.20.0.0/16	local	Active	No	Create Route Table

Both [Edit routes](#)

=> Connect Private Route Table with Private Subnet.

Internet Gateway (IGW)

igw-0658f6bcfb7b06589 / project-igw Actions

Details Info

Internet gateway ID
igw-0658f6bcfb7b06589

State
Attached

VPC ID
vpc-0a8f9035e38137dc0 | project-vpc

Owner
860668857234

Tags (1)

Key	Value
Name	project-igw

Manage tags

=> The IGW is attached to Public Route table.

NAT Gateway (NGW)

nat-0f8f154c46bfa6bc1 / project-nat Actions

Details

NAT gateway ID
nat-0f8f154c46bfa6bc1

NAT gateway ARN
arn:aws:ec2:us-east-1:860668857234:natgateway/nat-0f8f154c46bfa6bc1

VPC
vpc-0a8f9035e38137dc0 / project-vpc

Connectivity type
Public

Primary public IPv4 address
54.145.191.174

Subnet
subnet-03cbe4f9256aedfdb / public-web2-subnet

State
Available

Primary private IPv4 address
10.20.2.73

Created
Tuesday, July 28, 2026 at 08:42:58 GMT+5:30

State message Info
-

Primary network interface ID
eni-00a33ce9e54d4ecd2

Deleted
-

Secondary IPv4 addresses

Monitoring

Tags

=> The NGW is attached to the Private Route Table.

Security Groups

1. Security Group - In VPC

sg-0235ffe7671b837cf - project-vpc Actions

Details

Security group name
project-vpc

Owner
860668857234

Security group ID
sg-0235ffe7671b837cf

Inbound rules count
2 Permission entries

Description
ssh http

Outbound rules count
1 Permission entry

VPC ID
vpc-0a8f9035e38137dc0

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

☐

Name

Security group rule ID

IP version

Type

Protocol

Port range

Source

Description

☐

-

sgr-0eb418472f1b7a3c5

IPv4

HTTP

TCP

80

0.0.0.0/0

-

☐

-

sgr-0ae697fa48154cb6c

IPv4

SSH

TCP

22

0.0.0.0/0

-

=> Allowed ports are SSH (Port no: 22) and HTTP (Port no: 80)

2. Security Group- For the Database.

sg-023adef14984a1d22 - db-secg Actions

Details

Security group name db-secg	Security group ID sg-023adef14984a1d22	Description ssh mysql	VPC ID vpc-0a8f9035e38137dc0
Owner 860668857234	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2) Manage tags Edit inbound rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-05b36443366830377	IPv4	MySQL/Aurora	TCP	3306	0.0.0.0/0	-
☐	-	sgr-05aaceb29e86e2389	IPv4	SSH	TCP	22	0.0.0.0/0	-

=> Allowed ports are SSH (Port no: 22) and MYSQL (Port no: 3306)

Webservers & Database Server

Instances (3) Info Connect Instance state Actions Launch instances

Saved filter sets
Choose filter set

Instance state = running Clear filters

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Sei
☐	database-wbs	i-02e5534df569417f9	Running	t3.micro	3/3 checks passed	us-east-1d	-	-	-	-	disabled	db
☐	application-wbs	i-0b84e79adbbae3afb	Running	t3.micro	3/3 checks passed	us-east-1c	-	54.226.193.248	-	-	disabled	prc
☐	jump-server	i-032e75c3b6452a859	Running	t3.micro	3/3 checks passed	us-east-1a	-	3.230.125.107	-	-	disabled	prc

1. Web Server

- Jump Server
- Application Server

2. Database Server

=====

Application Webserver

=> Access it individually or through the Jump Server. => Install Nginx => Test.

Not Secure <https://54.226.193.248>

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to www.nginx.com.

Thank you for using nginx.

=> Install the packages:

```
root@application-wbs:~  
[root@application-wbs ~]# yum install -y php8.2 php-fpm php-mysqlnd php-pdo php-mbstring  
Last metadata expiration check: 0:55:25 ago on Tue Jul 28 15:58:07 2026.  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repository	Size
Installing:				
php8.2	x86_64	8.2.32-1.amzn2023.0.1	amazonlinux	9.6 k
php8.2-fpm	x86_64	8.2.32-1.amzn2023.0.1	amazonlinux	1.9 M
php8.2-mbstring	x86_64	8.2.32-1.amzn2023.0.1	amazonlinux	523 k
php8.2-mysqlnd	x86_64	8.2.32-1.amzn2023.0.1	amazonlinux	148 k
php8.2-pdo	x86_64	8.2.32-1.amzn2023.0.1	amazonlinux	87 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	97 k
apr-util-ldb	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	13 k
httpd-core	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.68-1.amzn2023.0.1	amazonlinux	12 k

=> Transfer files (html,css,php .etc) from your local machine to a virtual machine i.e. application web-server via FileZilla(Ubuntu or Mac) / WinSCP(Windows). Move the files using the following commands.

```
root@application-wbs:~  
[root@application-wbs ~]# mv /home/ec2-user/includes /usr/share/nginx/html  
[root@application-wbs ~]# mv /home/ec2-user/public /usr/share/nginx/html  
[root@application-wbs ~]# ls  
db.pem  
[root@application-wbs ~]# ls /usr/share/nginx/html  
404.html 50x.html icons includes index.html nginx-logo.png poweredby.png public  
[root@application-wbs ~]#
```

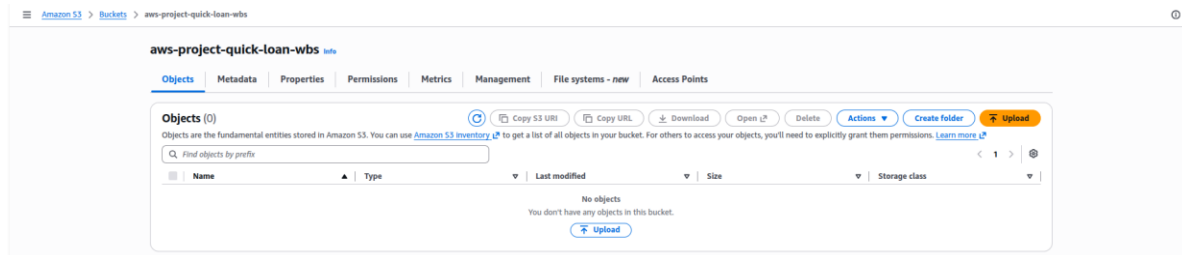
=> Update the Permissions.

```
root@application-wbs:~  
[root@application-wbs ~]# mv /home/ec2-user/includes /usr/share/nginx/html  
[root@application-wbs ~]# mv /home/ec2-user/public /usr/share/nginx/html  
[root@application-wbs ~]# ls  
db.pem  
[root@application-wbs ~]# ls /usr/share/nginx/html  
404.html 50x.html icons includes index.html nginx-logo.png poweredby.png public  
[root@application-wbs ~]# chown -R nginx:nginx /usr/share/nginx/html/public  
[root@application-wbs ~]# ls -l ^C  
[root@application-wbs ~]# ls -l /usr/share/nginx/html/public  
total 16  
-rw-rw-r--. 1 nginx nginx 1789 Jul 28 15:56 apply.php  
drwxrwxr-x. 2 nginx nginx 123 Jul 28 15:56 images  
drwxrwxr-x. 3 nginx nginx 42 Jul 28 15:56 includes  
-rw-rw-r--. 1 nginx nginx 3015 Jul 28 15:56 index.html  
-rw-rw-r--. 1 nginx nginx 2291 Jul 28 15:56 styles.css  
-rw-rw-r--. 1 nginx nginx 665 Jul 28 15:56 submit_application.php  
[root@application-wbs ~]# chmod -R 755 /usr/share/nginx/html/public  
[root@application-wbs ~]# ls -l /usr/share/nginx/html/public  
total 16  
-rwxr-xr-x. 1 nginx nginx 1789 Jul 28 15:56 apply.php  
drwxr-xr-x. 2 nginx nginx 123 Jul 28 15:56 images  
drwxr-xr-x. 3 nginx nginx 42 Jul 28 15:56 includes  
-rwxr-xr-x. 1 nginx nginx 3015 Jul 28 15:56 index.html  
-rwxr-xr-x. 1 nginx nginx 2291 Jul 28 15:56 styles.css  
-rwxr-xr-x. 1 nginx nginx 665 Jul 28 15:56 submit_application.php  
[root@application-wbs ~]#
```

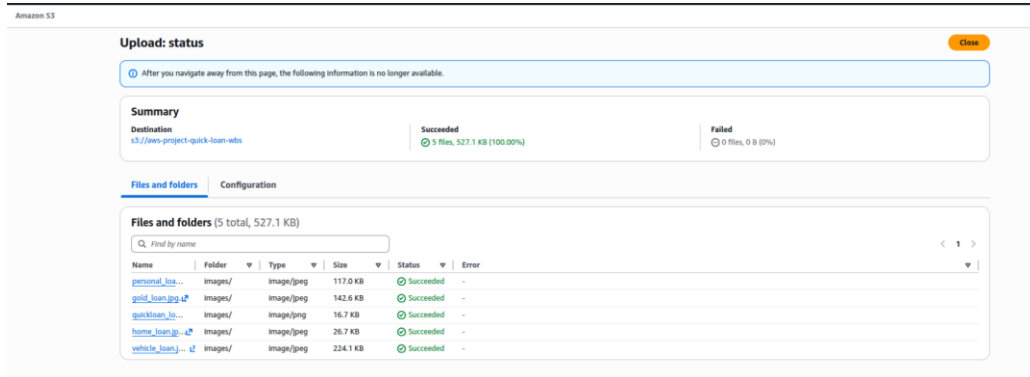
=====

S3 Bucket

=> Create S3 Bucket.



=> Upload the Folder.



=> After making changes in the index.html file using vim editor, proceed to connect the database, using 'vim <name of database connect>.php'

=> Replace endpoint, database username & database password.

=> Connect to the Database Server using its private-ip via Jump Server, then install the MariaDB package.

```
root@database-wbs:~#
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '?' for help. Type '?' to clear the current input statement.
MySQL [(none)]> CREATE DATABASE quickloan_db;
Query OK, 1 row affected (0.025 sec)

MySQL [(none)]> USE quickloan_db;
Database changed
MySQL [(none)]> CREATE TABLE applications (
  id INT AUTO INCREMENT PRIMARY KEY,
  name VARCHAR(255) NOT NULL,
  contact VARCHAR(20) NOT NULL,
  email VARCHAR(255) NOT NULL,
  loan_type VARCHAR(50) NOT NULL
);
Query OK, 0 rows affected (0.059 sec)

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| quickloan_db |
| sys |
+-----+
5 rows in set (0.001 sec)

MySQL [(none)]> desc applications;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| id | int | NO | PRI | NULL | auto_increment |
| name | varchar(255) | NO | | NULL | |
| contact | varchar(20) | NO | | NULL | |
| email | varchar(255) | NO | | NULL | |
| loan_type | varchar(50) | NO | | NULL | |
+-----+
5 rows in set (0.009 sec)

MySQL [(none)]>
```

ROUTE53

=> Create a hosted zone => Add the nameservers => Create a A-Record, Simple Routing.

► Hosted zone details

Records (3)

Accelerated recovery

DNSSEC signing

Hosted zone tags (0)

Records (3) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

TypeRouting p...Alias

☐	Record ... ▼	Type ▼	Routin... ▼	Differ... ▼	Alias ▼	Value/Route traffic to ▼	TTL (s... ▼	Health ... ▼	Evaluat... ▼	Record ID
☐	Jonathanp...	A	Simple	-	No	54.226.193.248	300	-	-	-
☐	Jonathanp...	NS	Simple	-	No	ns-939.awsdns-53.net. ns-1039.awsdns-01.org. ns-1608.awsdns-09.co.uk. ns-185.awsdns-23.com.	172,800	-	-	-
☐	Jonathanp...	SOA	Simple	-	No	ns-939.awsdns-53.net. awsd...	900	-	-	-

=> Access the website via the domain.

Welcome to nginx!QuickLoan - Your Trusted Fin...Apply for Loan - QuickLoan

Not Securehttp://jonathanpopat.com

Private browsing

Apply for Loan

Your Financial Dreams, Realized.

QuickLoan provides fast, reliable, and affordable financial solutions.

Our Loan Products



Home Loans
Own your dream home with our flexible home loan options.



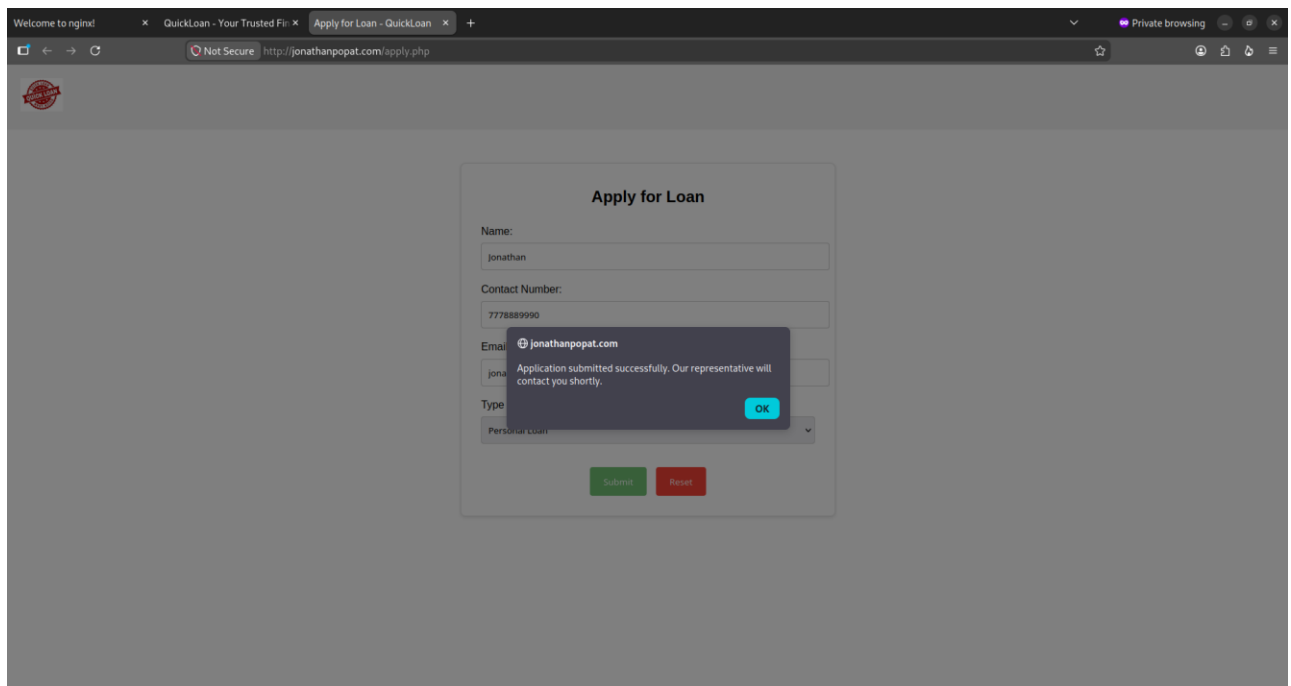
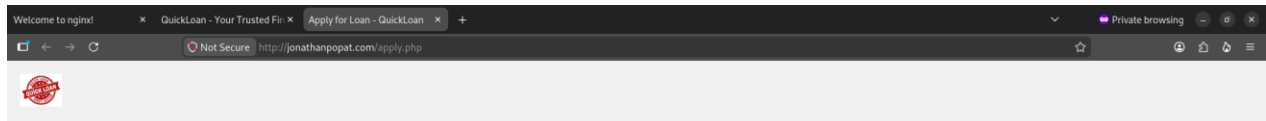
Gold Loans
Get instant cash with our hassle-free gold loan services.



Vehicle Loans
Drive your dream vehicle with our easy vehicle loan plans.



Personal Loans
Meet your urgent financial needs with our quick personal loans.



```
root@database-wbs:~  
| quickloan_db |  
| sys |  
+-----+  
5 rows in set (0.001 sec)  
  
MySQL [(none)]> select * from applications;  
ERROR 1046 (3D000): No database selected  
MySQL [(none)]> USE quickloan_db;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A  
  
Database changed  
MySQL [quickloan_db]> SHOW TABLES;  
+-----+  
| Tables_in_quickloan_db |  
+-----+  
| applications |  
+-----+  
1 row in set (0.001 sec)  
  
MySQL [quickloan_db]> SELECT * FROM applications;  
+-----+  
| id | name | contact | email | loan_type |  
+-----+  
| 1 | Jonathan | 7778889990 | jonathan@123 | Personal Loan |  
| 2 | Jonathan | 7778889990 | jonathanp@123 | Personal Loan |  
+-----+  
2 rows in set (0.001 sec)  
  
MySQL [quickloan_db]> 
```

Auto Scaling Groups

=> Create an Image

Amazon Machine Images (AMIs) (1) Info

Owned by me Find AMI by attribute or tag

☐	Name	AMI name	AMI ID	Source	Owner	Visibility	Status	Creation date	Platform
☐	application-web-image		ami-0d7c261f01a0ba516	860668857234/application-web-image	860668857234	Private	Available	2026/07/29 01:32 GMT+5:30	Linux/UNIX

=>Create Target Group

Target groups (1/1) Info What's new

Filter target groups

☒	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☒	aws-wbs-tg	arn:aws:elasticloadbalancing...	80	HTTP	Instance	aws-wbs-lb	vpc-0a8f9035e38137dc0

=> Create Load Balancer

aws-wbs-lb

▼ Details

Load balancer type

Application

Scheme

Internet-facing

Status

ⓘ Provisioning

VPC

vpc-0a8f9035e38137dc0 ⓘ

Load balancer IP address type

IPv4

Availability Zones

subnet-02d3639adc4f668d3 ⓘ us-east-1b (use1-az2)

subnet-03cbe4f9256aeefdb ⓘ us-east-1c (use1-az4)

Date created

July 29, 2026, 01:55 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:860668857234:loadbalancer/app/aws-wbs-lb/38b26f37fa04146b

DNS name info

aws-wbs-lb-1597869534.us-east-1.elb.amazonaws.com [A Record]

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1) ⓘ

Manage rules ▼

Manage listener ▼

Add listener

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

Filter listeners

☐

Protocol:Port

☐

HTTP:80

Default action

• Forward to target group

aws-wbs-tg ⓘ: 1 (100%)

Target group stickiness: Off

Rules

1 rule

ARN

Not applicable

Security policy

Not applicable

Default SSL/TLS certificate

Not applicable

mTLS

Not applicable

Trust store

Not applicable

=> Launch Template

Launch templates (1/1) ⓘ

Actions

Create launch template

Search

☒

Launch template ID

☒

Launch template name

☐

Default version

☐

Latest version

☐

Create time

☐

Created by

☐

Managed

☐

Operator

lt-04d74b425c454fdde

aws-wbs-template

1

1

2026-07-28T20:29:49.000Z

arn:aws:iam:860668857234:r...

false

-

=> Auto Scaling Groups

Auto Scaling groups (1/1) ⓘ

Launch configurations

Launch templates ⓘ

Actions

Create Auto Scaling group

Search your Auto Scaling groups

☒

Name

☐

Status - new

☐

Launch template/configuration ⓘ

☐

Instances

☐

Instance health - new

☐

Desired capacity

☐

Min

☐

Max

☐

Availability Zones

☐

Creation time

aws-wbs-autoscale

Updating capacity

aws-wbs-template | Version Default

0

-

2

2

5

2 Availability Zones

Wed Jul 29 2026 02:06:58 GMT+05:30 (India Stand...

Instances (6) ⓘ

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

☐

Name ⓘ

☐

Instance ID

☐

Instance state

☐

Instance type

☐

Status check

☐

Availability Zone

☐

Public IPv4 DNS

☐

Public IPv4 ...

☐

Elastic IP

☐

IPv6 IPs

☐

Monitoring

☐

Sei

database-wbs

i-02e3534df369417f9

Running

t3.micro

3/3 checks passec

us-east-1d

-

-

-

disabled

db

i-02f09913bdba7679e

Running

t3.micro

ⓘ Initializing

us-east-1c

-

23.22.207.145

-

-

disabled

prc

☐

public-web2

i-062cf0ca90848fdd

Running

t3.micro

3/3 checks passec

us-east-1c

-

100.31.2.5

-

-

disabled

prc

☐

application-wbs

i-0b84e79adbaae3afb

Running

t3.micro

3/3 checks passec

us-east-1c

-

54.226.193.248

-

-

disabled

prc

☐

i-0c4077d7b6a9ed631

Running

t3.micro

ⓘ Initializing

us-east-1b

-

44.204.34.58

-

-

disabled

prc

☐

jump-server

i-032e75c3b6452a859

Running

t3.micro

3/3 checks passec

us-east-1a

-

3.230.125.107

-

-

disabled

prc

=> 4 previous running; 2 new formed (desired)

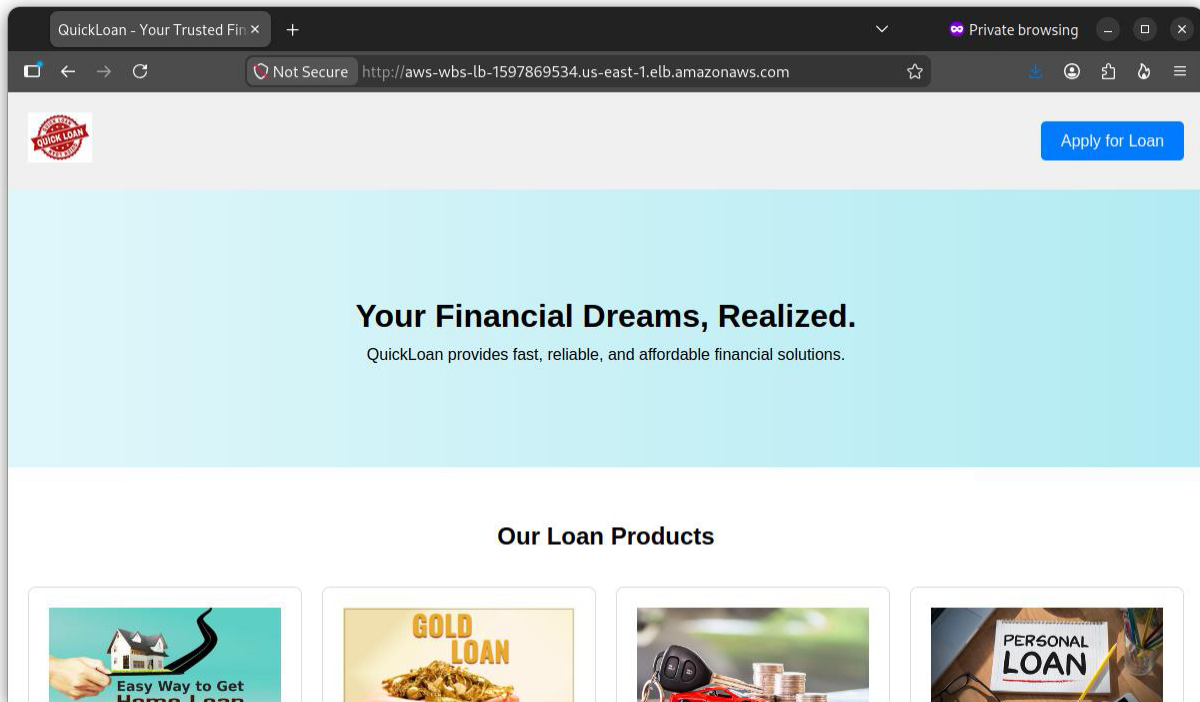
=====

ROUTE53

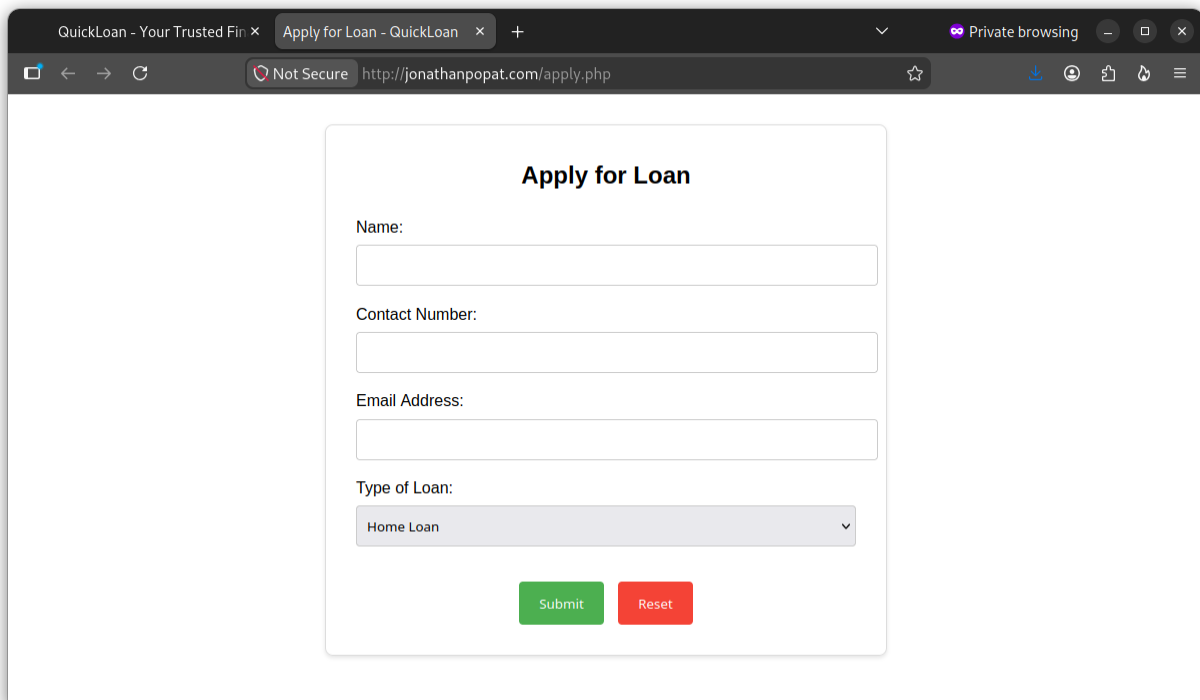
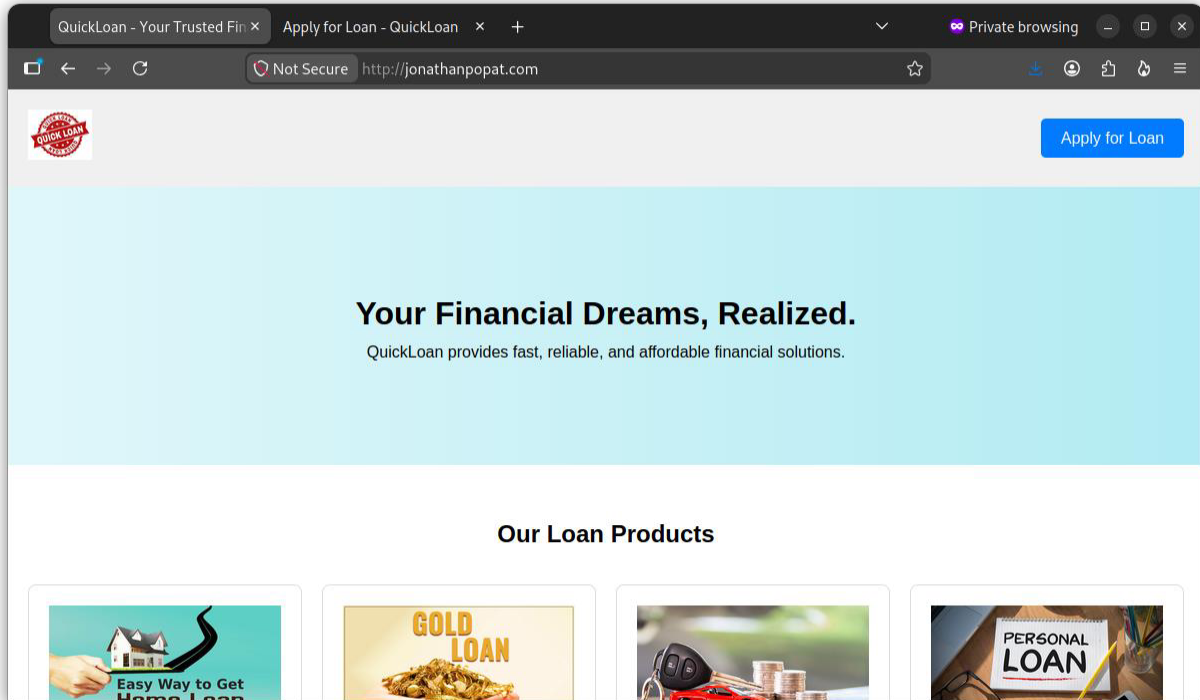
=> Attaching the Load balancer to Route53

=> By Creating CName Record

=> test = accessible through load-balancer dns name



=> Load-Balancer attached to domain



Conclusion: The QuickLoan project successfully demonstrates the development of a simple and user-friendly online loan application system. It enables users to explore different loan products and submit applications conveniently through a web interface. The integration of PHP and MySQL allows application data to be processed and stored efficiently. Overall, the project provides a basic foundation for a digital loan application platform and can be further enhanced with features such as user authentication, application tracking, and improved security.